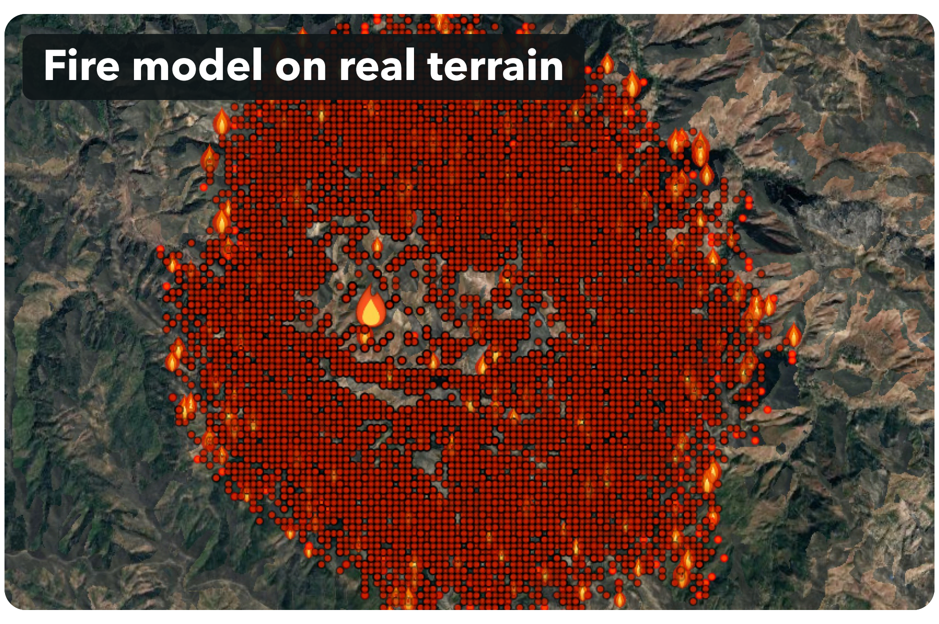
How the dog decides

Simulator views on real terrain. Robot image: design concept.

① Predict

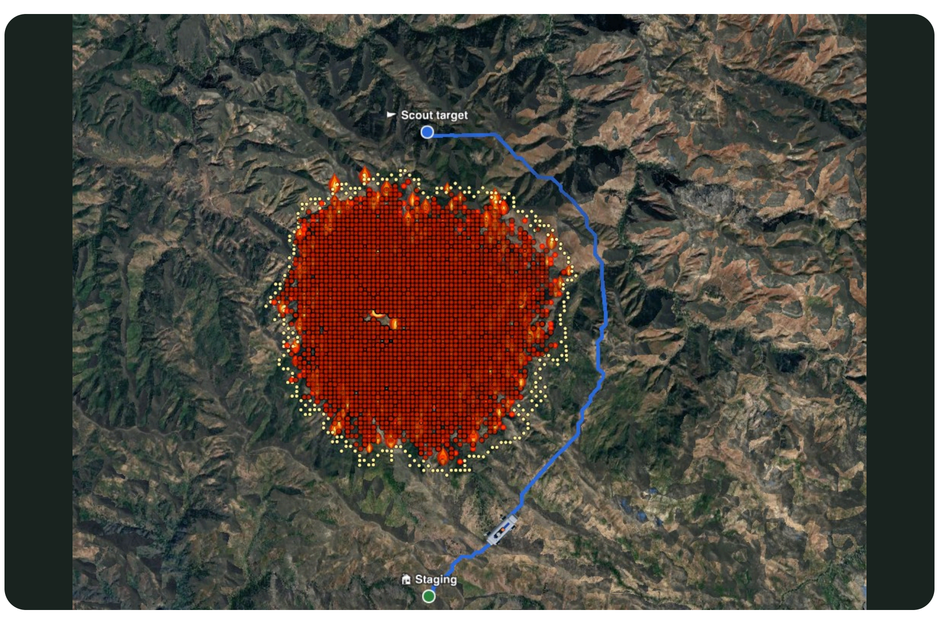


Fire model on real terrain

50 m cells · **15 min** forecast steps

Forecasts when each 50 m cell will burn.

② Safe route



Blue: route to scout target

Stays 30+ min ahead of the forecast fire.

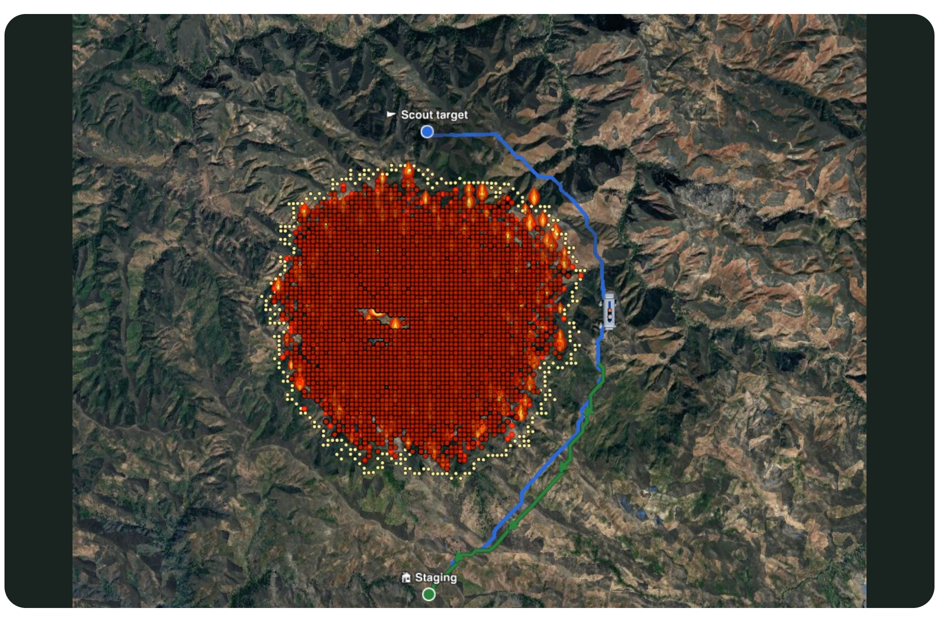
3D On real ground

Slopes + land cover affect speed

Open **1.0×** · Forest **0.7×** · Brush **0.5×**

Uphill is slower. No water or slopes over 35°.

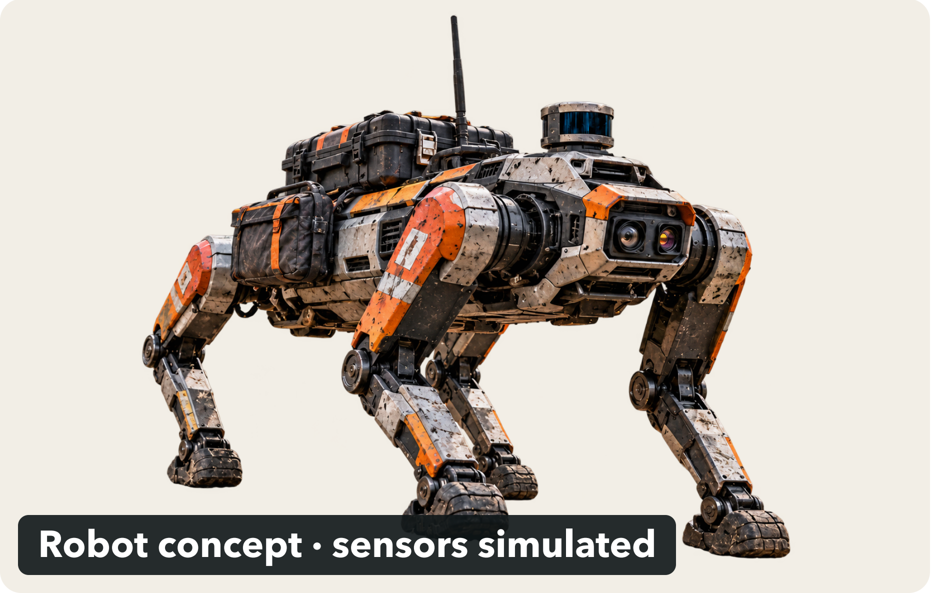
③ Escape path



Green: escape path home

Aims for a 60 min clear way home.

④ Scout

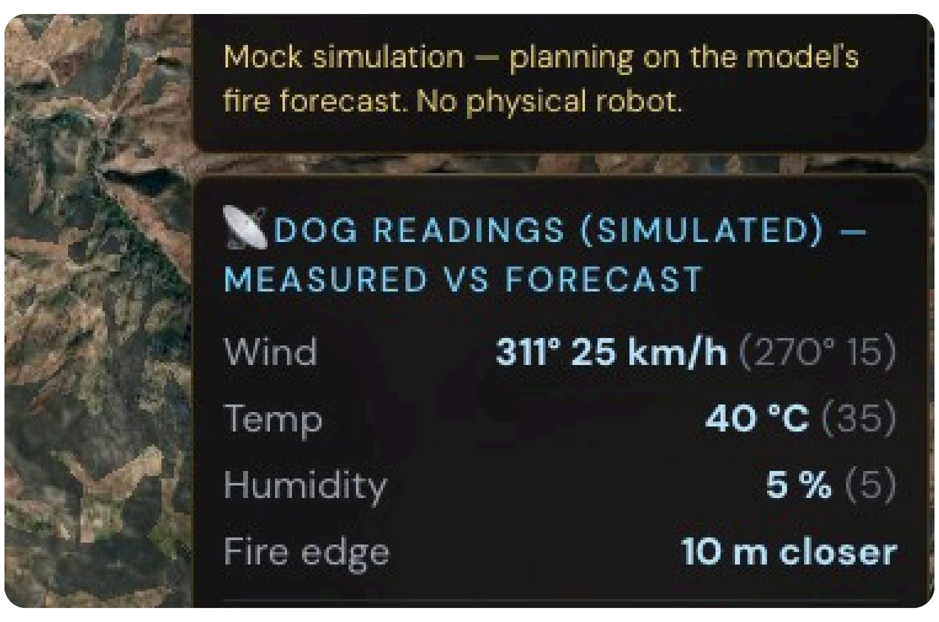


Robot concept · sensors simulated

Wind · temperature · humidity · fire edge

Measures what the forecast only guessed.

⑤ Feedback



Mock simulation — planning on the model's fire forecast. No physical robot.

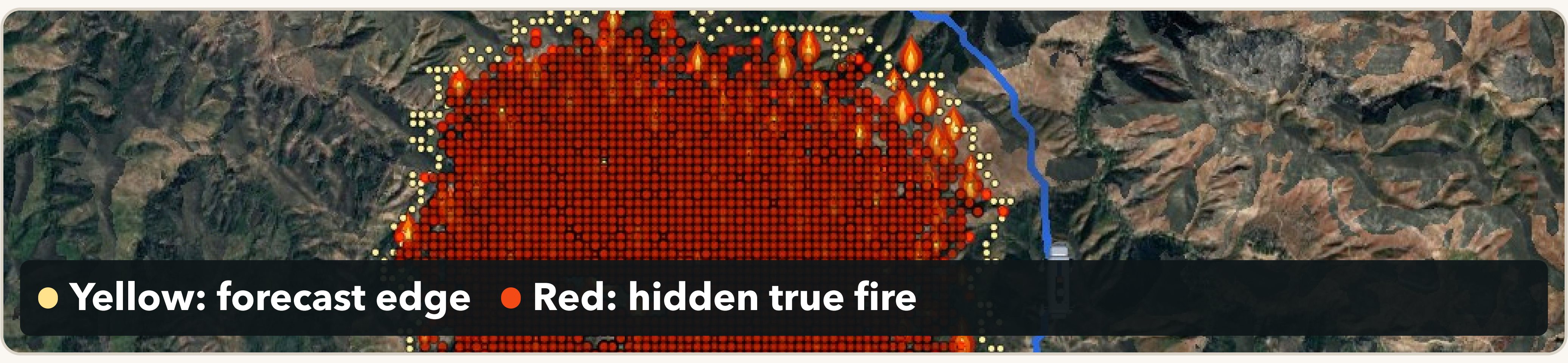
DOG READINGS (SIMULATED) — MEASURED VS FORECAST

Wind	**311° 25 km/h** (270° 15)
Temp	**40 °C** (35)
Humidity	**5 %** (5)
Fire edge	**10 m closer**

Launch **82%** · Shift **61%** · Report **80%**

The dog's readings improve the next forecast.

How we tested it



● Yellow: forecast edge ● Red: hidden true fire

Forecast fire
What the dog plans on

⇠ dog's readings

True fire (hidden)
Hotter, drier, wind shift

Mock simulation; no physical robot. The dog learns about the hidden true fire through simulated sensor reports.

Status & next

- ✔ ML fire forecast trained and tested
- ✔ 3D browser simulator on real terrain
- ✔ Containment planner (search)
- ✔ Robot-dog loop (route, escape, scout, feedback) in mock simulation
- → Run forecast + planner on the dog or its controller
- → Real sensors feeding the forecast
- → Scout the planned containment line
- → Tests on real robot dogs

Open-source code + demo video

Data & references

Code: github.com/sxz2012/wildfire-simulation

NASA FIRMS (VIIRS active-fire detections) · Open-Meteo weather archive · ESA WorldCover 10 m land cover · Cesium World Terrain · Google Photorealistic 3D Tiles

Rothermel, R. C. (1972). *A mathematical model for predicting fire spread in wildland fuels.* USDA Forest Service INT-115.

Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD '16.*